Probabilistic Artificial Intelligence

Programming Assignment 2

Bayesian Deep Learning – Modeling and Analyzing Uncertainty

Submitted By: Wishal Fatima

1. Model Descriptions

MC-Dropout Approximates Bayesian Inference

A standard neural network outputs a single point prediction with no notion of how confident it is. Dropout is normally used only during training, as a regularizer that randomly zeroes a fraction of units on each forward pass. Gal and Ghahramani (2016) showed that a network trained with dropout before every weight layer can be reinterpreted as an approximation to a deep Gaussian Process, and that keeping dropout active at test time turns ordinary dropout into an approximate Bayesian inference procedure.

Intuition:

In MC-Dropout, each forward pass uses a different random dropout mask, creating a different version of the neural network. Running the model multiple times produces a set of predictions that approximate the Bayesian prediction distribution. The average of these predictions gives the final prediction, while the variation between predictions represents epistemic uncertainty.

My Implementation:

MCDropoutClassifier and MCDropoutRegressor contain dropout layers between hidden layers. At evaluation time, the model is kept in model.train() mode so dropout masks remain active rather than being disabled. I run num_passes = 50 stochastic forward passes per input and aggregate the resulting predictions to obtain a predictive mean and variance. For regression, the network predict both (mean and variance), so each pass already produces an aleatoric variance estimate; the spread across passes gives the epistemic component.

Deep Ensembles Approximate Bayesian Inference

Deep Ensembles provide an alternative approach for uncertainty estimation by training multiple independently initialized neural networks. Deep Ensembles (Lakshminarayanan et al., 2017) take a non-Bayesian-in-the-strict-sense but practically effective route to the same goal: obtaining a distribution over predictions rather than a single point estimate.

Intuition:

Deep Ensembles approximate Bayesian uncertainty by training multiple independently initialized models with the same architecture. Since each model learns a slightly different solution, the collection of models represents different possible predictions. The average prediction gives the final output, while the disagreement between models measures epistemic uncertainty.

My Implementation:

I independently initialized k = 5 models and trained for both the classification task and the regression task. Each ensemble member uses the same architecture but starts from different random weights (the global seed advances between model constructions, so no manual per-model seeding was required).

For regression, each member also predicts its own (input-dependent) variance, so the ensemble's total predictive variance combines

(a) the average of each member's own variance and

(b) the variance of the means across members — same decomposition logic as MC-Dropout.

Predictions are combined using ensemble_predict_classifier / ensemble_predict_regressor.

Side-by-Side Comparison

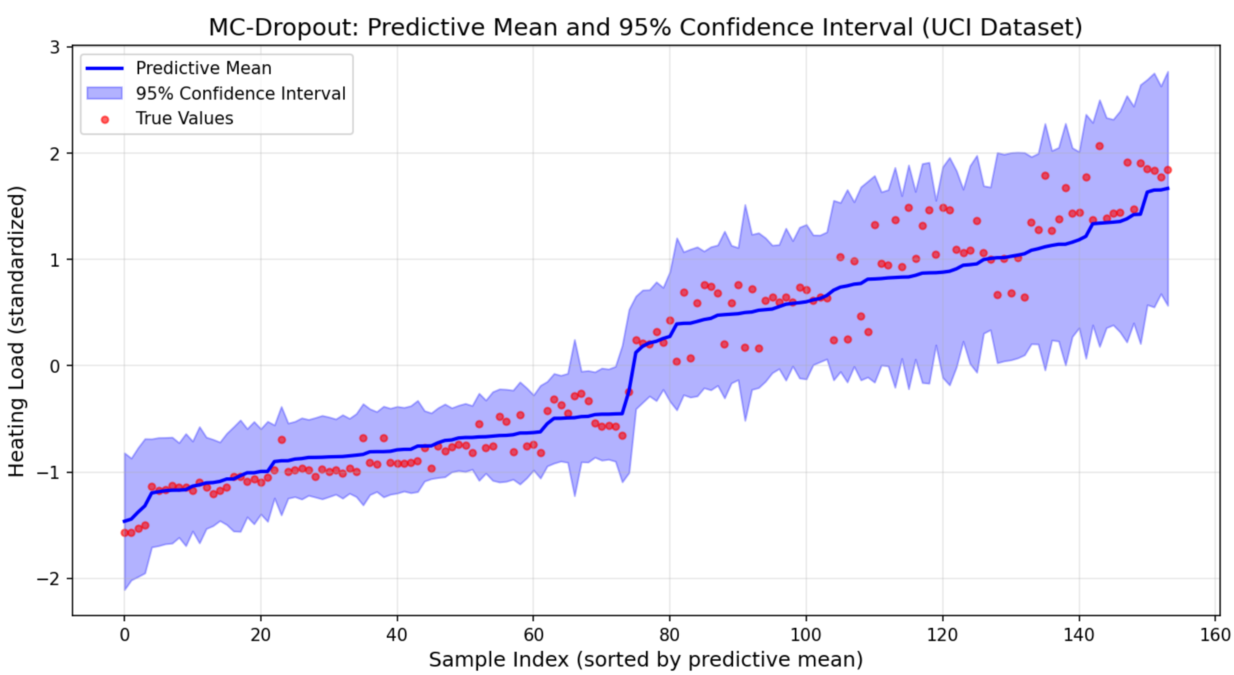
Aspect	MC-Dropout	Deep Ensembles
Number of trained models	1	k = 5
Source of stochasticity	Random dropout masks at inference	Random weight initialization at training
Inference cost	High - 50 stochastic forward passes per input	Moderate - 5 forward passes per input (one per member)
Training cost	Low - 1 model trained once	High - 5 models trained independently (~5×)
Uncertainty estimate	Variance across stochastic passes of one model	Variance across independently trained models
Theoretical grounding	Approximate variational Bayesian inference (Gal & Ghahramani, 2016)	Not strictly Bayesian, but a strong empirical approximation (Lakshminarayanan et al., 2017)

Below are my findings:

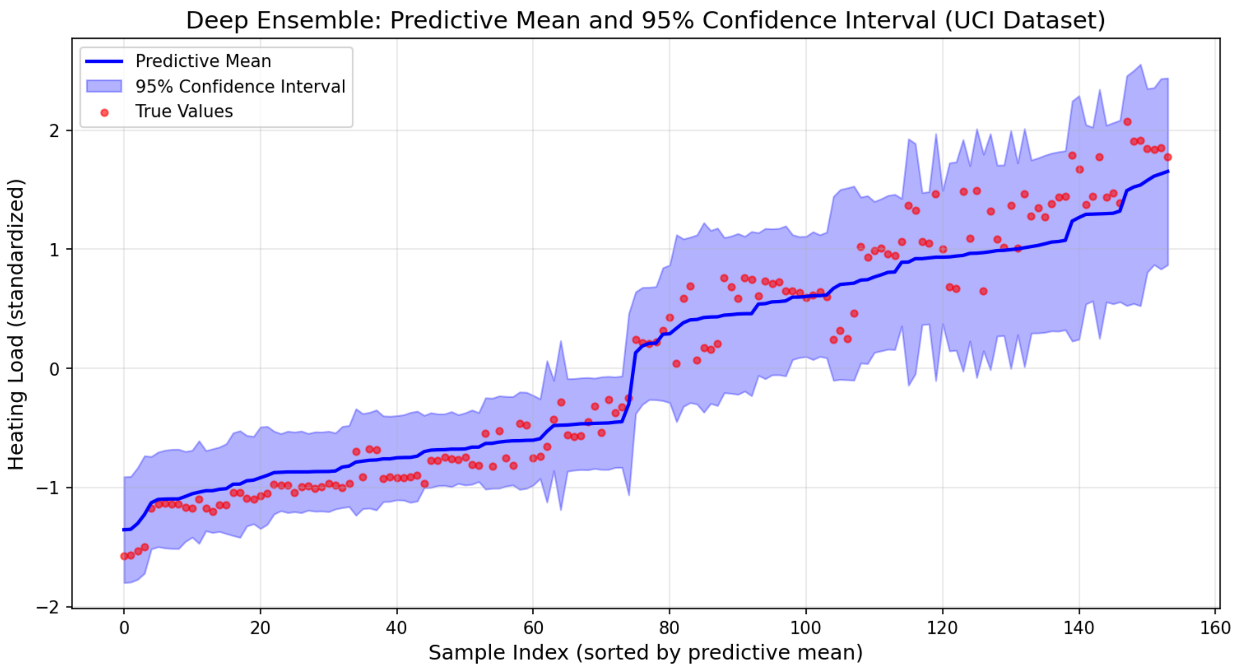
(All values in this comparison are in standardized units)

Metric	**MC Dropout(UCI)**	**Deep Ensemble(UCI)**
MC passes / Ensemble models	50 passes	5 models
Test Set (UCI)		
RMSE	2.3853	2.2127
NLL	-1.0196	-1.0933
ECE	0.063077	0.041903
Training Set		
Training RMSE	2.1827	2.0320
Training NLL	-1.0656	-1.1390

MC-Dropout on UCI



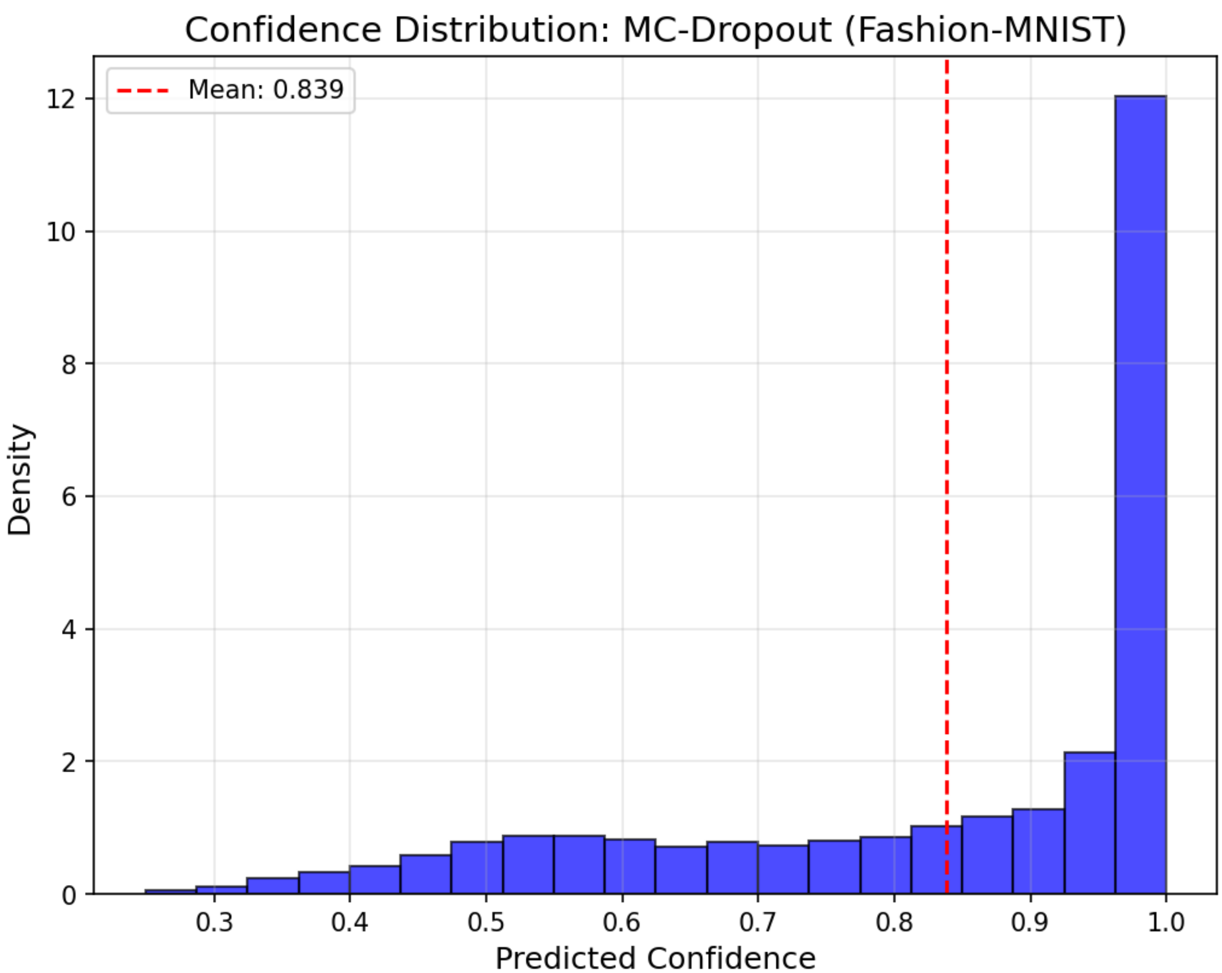
Deep Ensemble on UCI

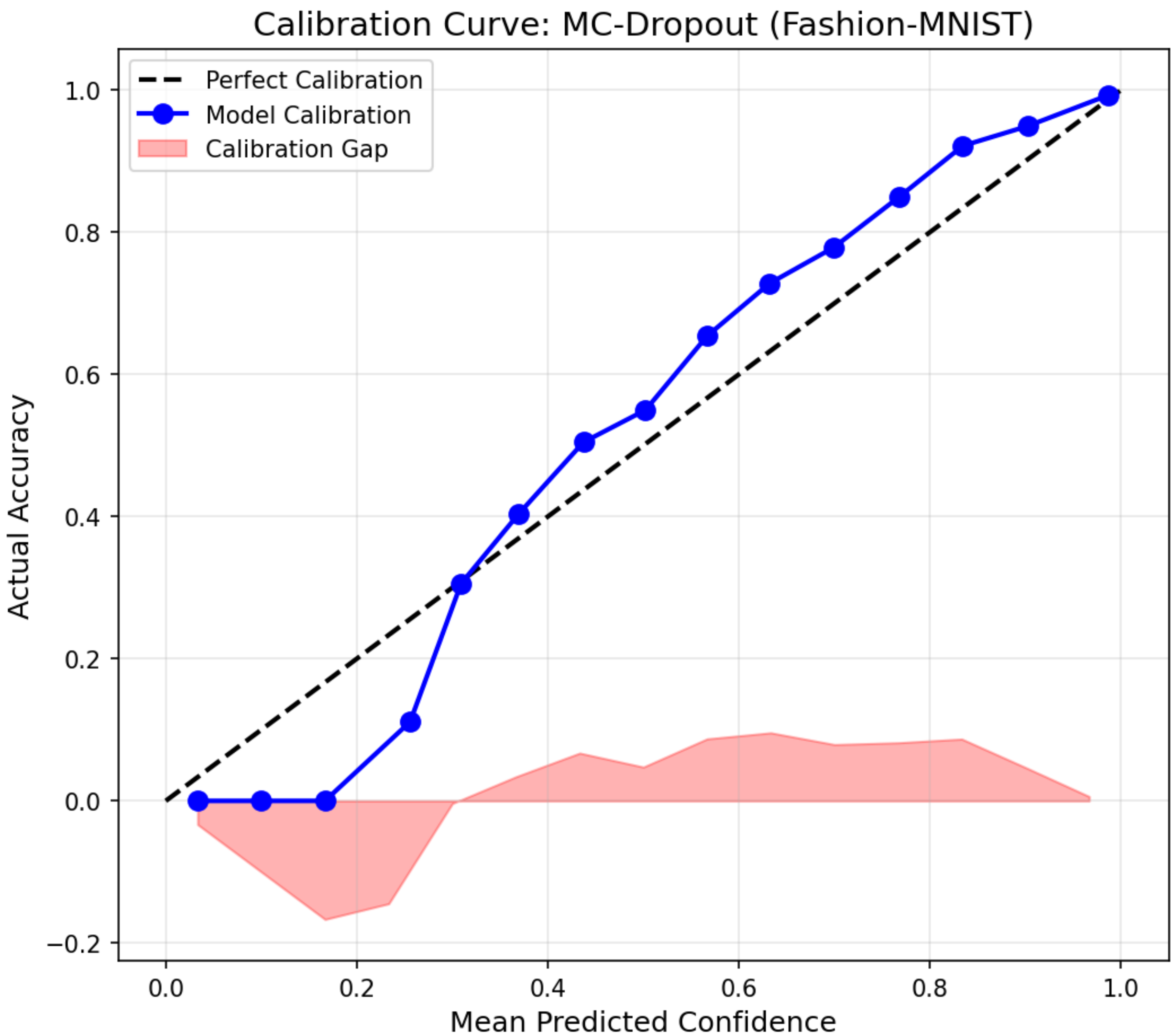


Metric	**MC Dropout (Fashion-MNIST)**	**Deep Ensemble (Fashion-MNIST)**
MC passes / Ensemble models	50 passes	5 models
Test Set (UCI)		
Accuracy	87.55%	88.12%
NLL	0.3520	0.3366
ECE	0.0371	0.0277
Training Set		
Training Accuracy	89.43%	89.97%

Training NLL	0.3021	0.2794

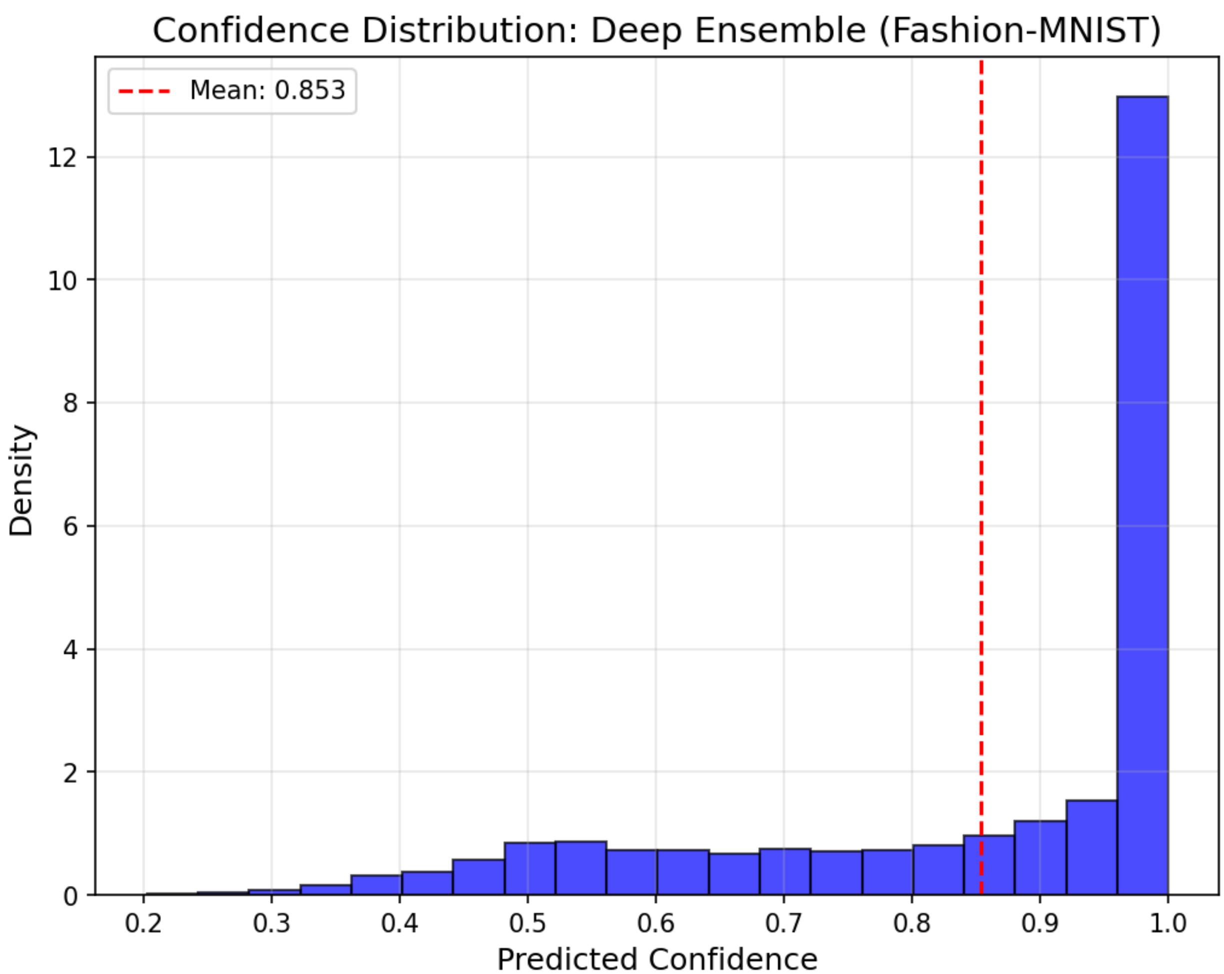
MC-Dropout on Fashion-MNIST



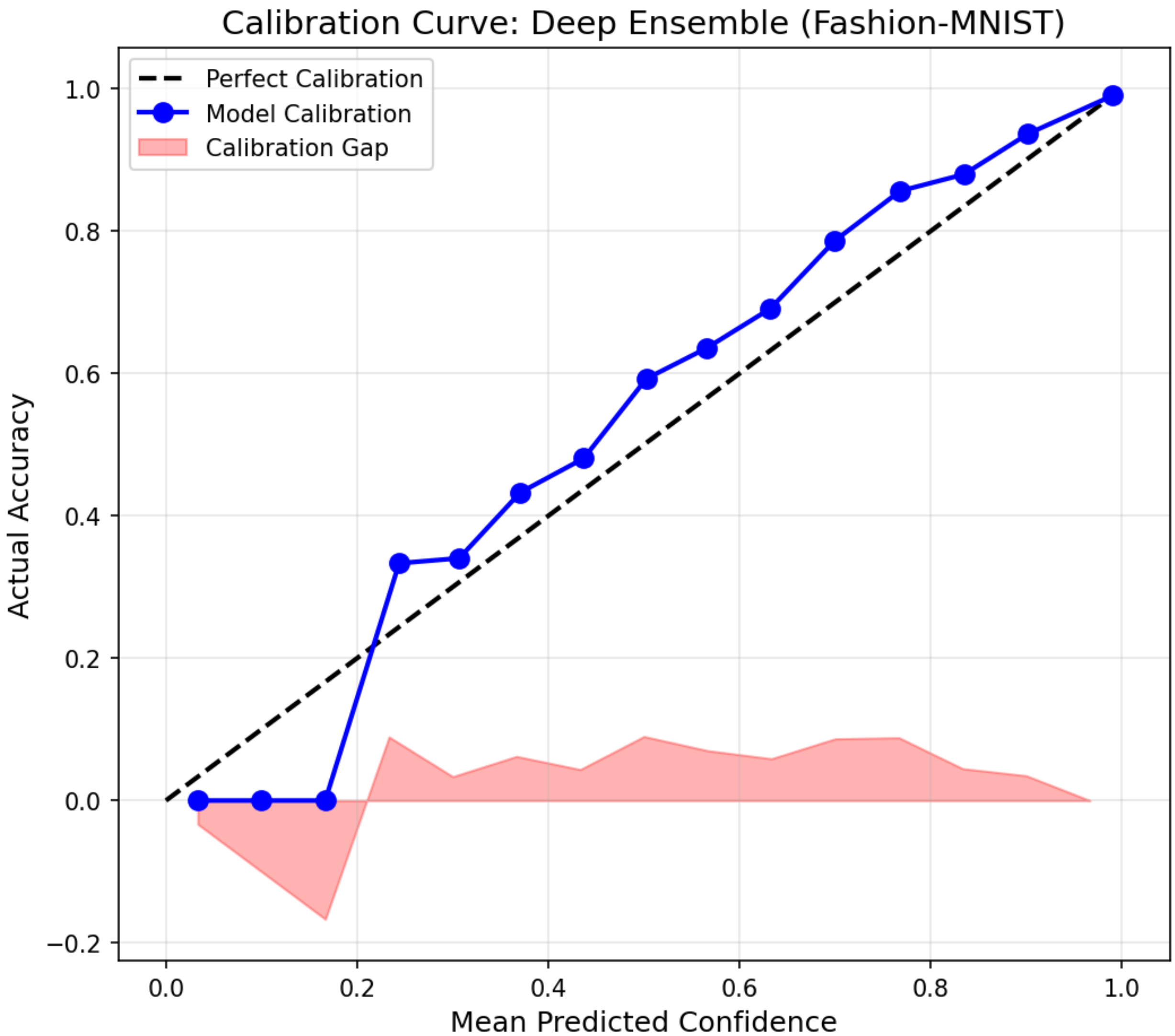


Calibration Curve: MC-Dropout (Fashion-MNIST)
Perfect Calibration
Model Calibration
Calibration Gap
Actual Accuracy
Mean Predicted Confidence
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0

Deep Ensemble on Fashion-MNIST

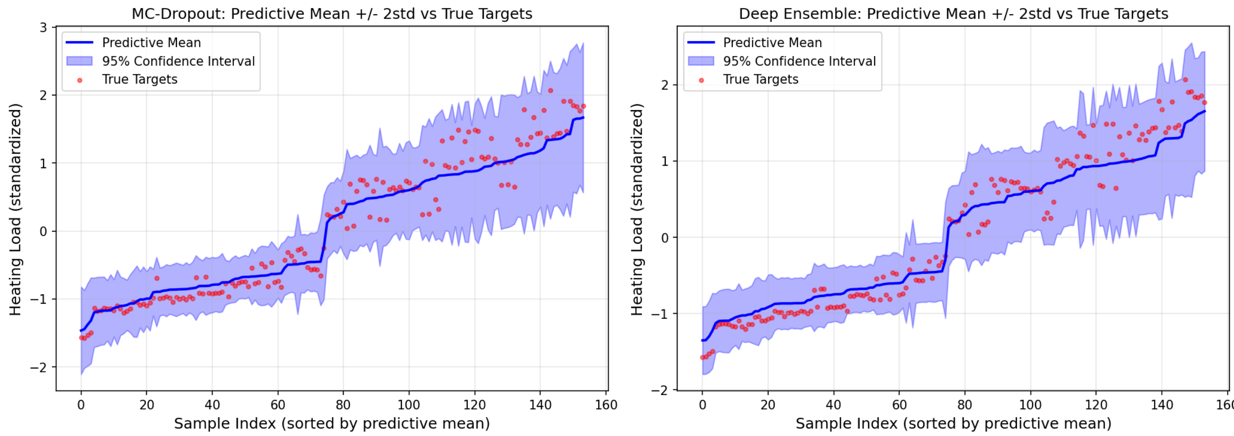


Deep Ensemble on Fashion-MNIST



2. Evaluation summary:

Regression Uncertainty (UCI Energy Efficiency)



Comparison between Models

Metric	MC-Dropout	Deep Ensemble
Uncertainty Components		
Mean uncertainty	0.6434	0.5626
Mean absolute error	0.1780	0.1801

Deep Ensemble produces slightly lower uncertainty (0.5626) than MC-Dropout (0.6434). Both models achieve almost identical prediction error, with MAE of 0.1780 for MC-Dropout and 0.1801 for Deep Ensemble. MC-Dropout is slightly more conservative because it estimates larger uncertainty, while Deep Ensemble gives more stable predictions because it averages multiple independently trained models.

Correlation between Uncertainty and Error

MC-Dropout: Correlation between uncertainty and error = 0.6376
Deep Ensemble: Correlation between uncertainty and error = 0.7204

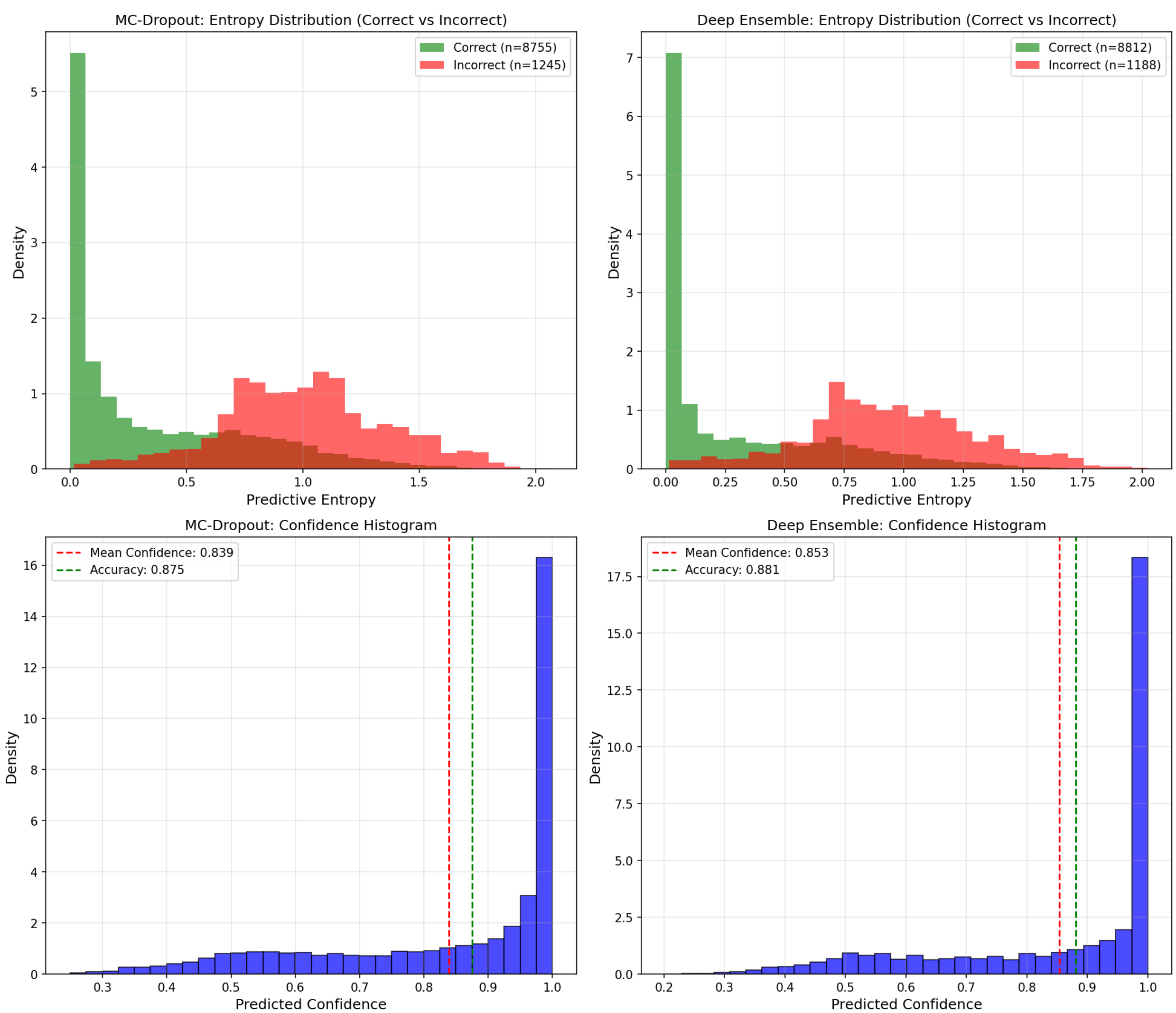
Where is uncertainty high or low?

Regression uncertainty was analyzed by comparing predictive means and confidence intervals with the true heating load values. The uncertainty was higher in regions with larger prediction variability, especially for extreme heating-load values and transition regions where the model prediction changed significantly. In these regions, the average uncertainty (2×std) reaches 1.04, more than three times higher than the low uncertainty region (0.36).These areas likely contain less frequently observed building configurations, causing higher epistemic uncertainty.

Low uncertainty was observed in regions with typical heating-load values where the model had learned stable patterns from the training data. The narrow confidence intervals indicate consistent predictions across stochastic forward passes or ensemble members.

Both MC-Dropout and Deep Ensemble methods produced meaningful uncertainty estimates, as shown by the positive correlation between uncertainty and prediction error. Deep Ensemble achieved a stronger uncertainty-error correlation (0.7204) compared to MC-Dropout (0.6376), indicating slightly better uncertainty calibration. However, MC-Dropout produced similar predictive accuracy while requiring less computational cost.

Classification Uncertainty (Fashion-MNIST)



	MC Dropout Results	Deep Ensemble Results
Total test samples	10000	10000
Correct predictions	8755 (87.55%)	8812 (88.12%)
Incorrect predictions	1245 (12.45%)	1188 (11.88%)
Mean entropy (correct)	0.3543	0.2998
Mean entropy (incorrect)	1.0059	0.9387
Mean confidence	0.8387	0.8535
Overall accuracy	0.8755	0.8812
Calibration gap	0.0368	0.0277
Overall mean entropy	0.4355	0.3757
Entropy Gap (Incorrect - Correct)	0.6516	0.6389

Entropy

Correct predictions (green) are heavily concentrated at very low predictive entropy, with mean entropy of 0.3543 (MC-Dropout) and 0.2998 (Deep Ensemble). The model is "sure" almost exactly when it is right.

Incorrect predictions (red) are spread across a much wider band of higher entropy values, with mean entropy of 1.0059 (MC-Dropout) and 0.9387 (Deep Ensemble). Very few incorrect predictions occur at near-zero entropy.

The gap of 0.65 for MC-Dropout and 0.64 for Deep Ensemble indicates that both models are significantly more uncertain when they make incorrect predictions. This is desirable because it means the model's uncertainty estimates are meaningful. The statistical test confirms this with $p < 0.001$ for both models (MC-Dropout: 0.000000e+00, Deep Ensemble: 0.000000e+00), confirming that predictive entropy is a genuinely useful signal for flagging likely-wrong predictions, not just visual noise.

MC-Dropout shows a slightly larger raw entropy range (0.4355 vs 0.3757), consistent with its generally noisier, higher-variance uncertainty estimates relative to the ensemble's smoother disagreement-based estimate.

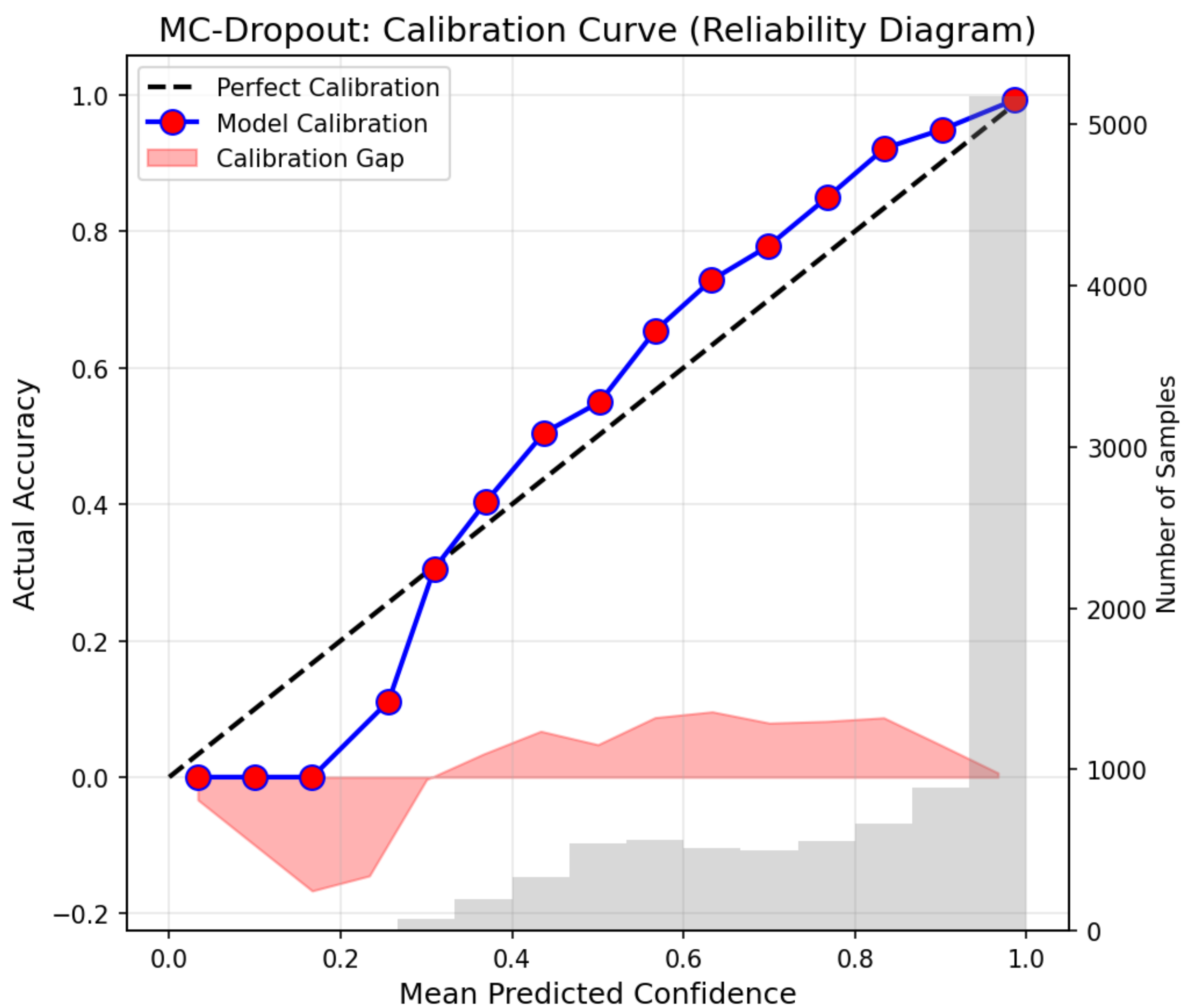
Confidence histograms

Both models concentrate the large majority of predictions near confidence = 1.0, which is expected given most Fashion-MNIST test images are easy to classify correctly.

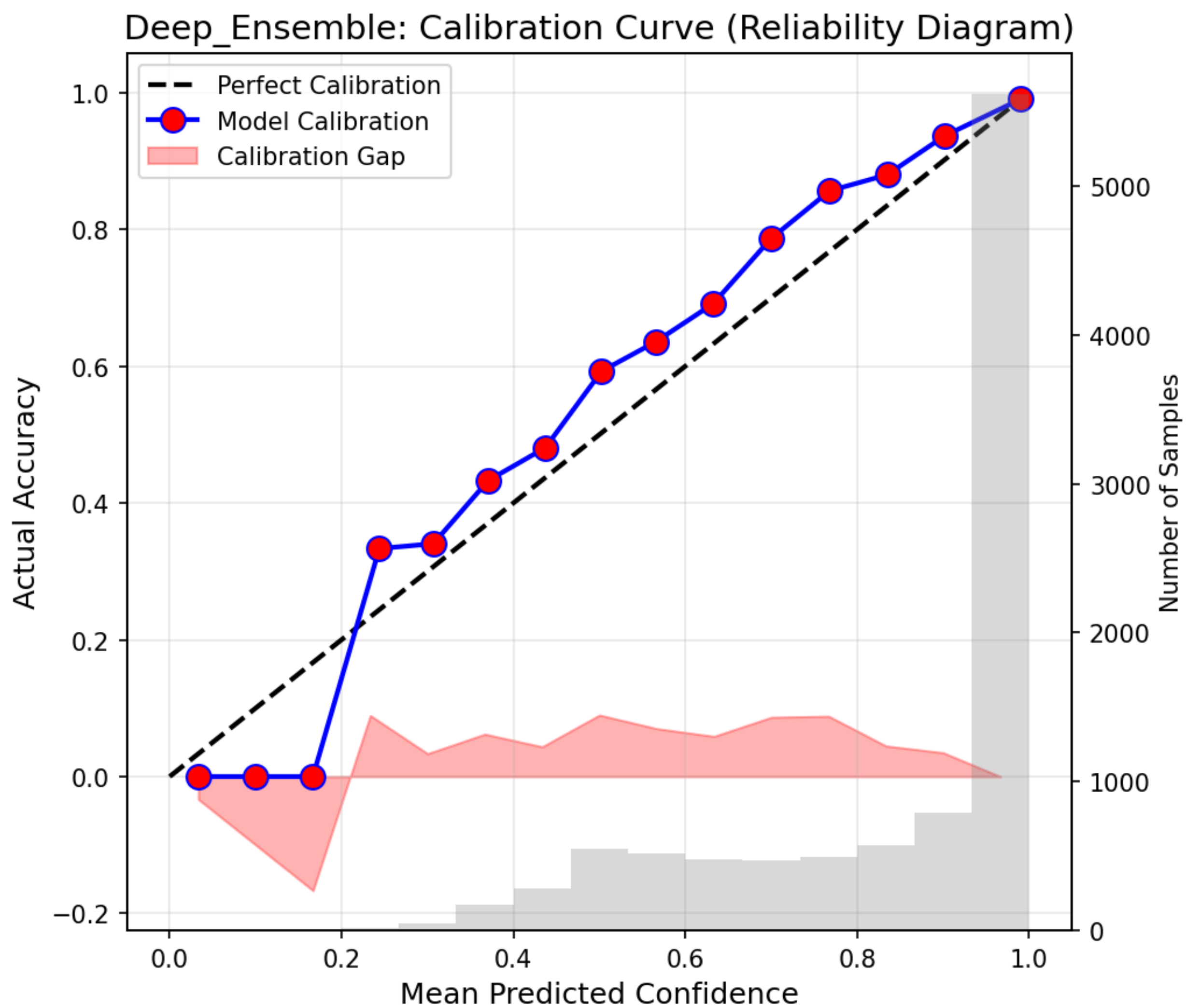
Mean confidence (0.839 for MC-Dropout, 0.853 for Deep Ensemble) sits slightly below the corresponding accuracy (0.875 / 0.881), both models are mildly under confident on average rather than overconfident, which is the safer direction to choose on for most applications.

Both models are well-calibrated with small calibration gaps (< 0.04). The Deep Ensemble shows slightly better calibration with a gap of 0.0277 compared to MC-Dropout's 0.0368.

Calibration curves (Reliability diagrams)



Calibration analysis for MC-Dropout:
Overconfident bins (confidence > accuracy): 2 bins
Underconfident bins (confidence < accuracy): 10 bins



Calibration analysis for Deep_Ensemble:
Overconfident bins (confidence > accuracy): 1 bins
Underconfident bins (confidence < accuracy): 11 bins

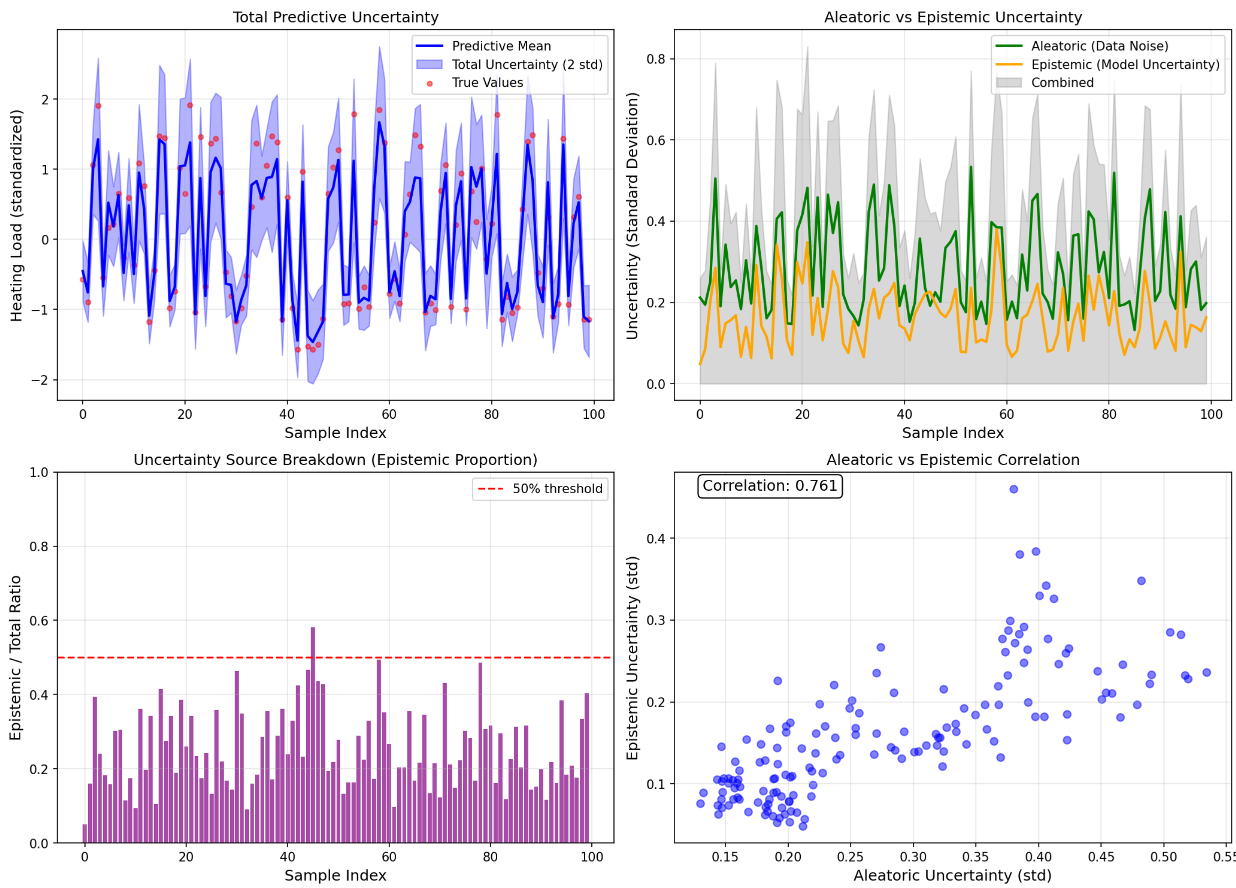
Both curves hug the diagonal “perfect calibration” line closely for confidence above ~0.3, where the bulk of the samples lie (see the grey sample-count bars).

The lowest three confidence bins show 0% accuracy for both models, but the grey histogram shows these bins contain very few samples, so they contribute little to the overall ECE and should not be over-interpreted.
Deep Ensemble’s curve is marginally closer to the diagonal overall, consistent with its lower ECE (0.0277 vs. MC-Dropout's 0.0371).
Both models are predominantly underconfident across the mid-to-high confidence range (the calibration-gap shading sits above the diagonal in most bins), again indicating a conservative rather than overconfident bias.
Both ECE values are below 0.05, indicating good calibration. The Deep Ensemble is better calibrated. Both models have more underconfident bins (11 for MC-Dropout, 10 for Deep Ensemble) than overconfident bins (2 for both), indicating they are safely cautious rather than dangerously overconfident.

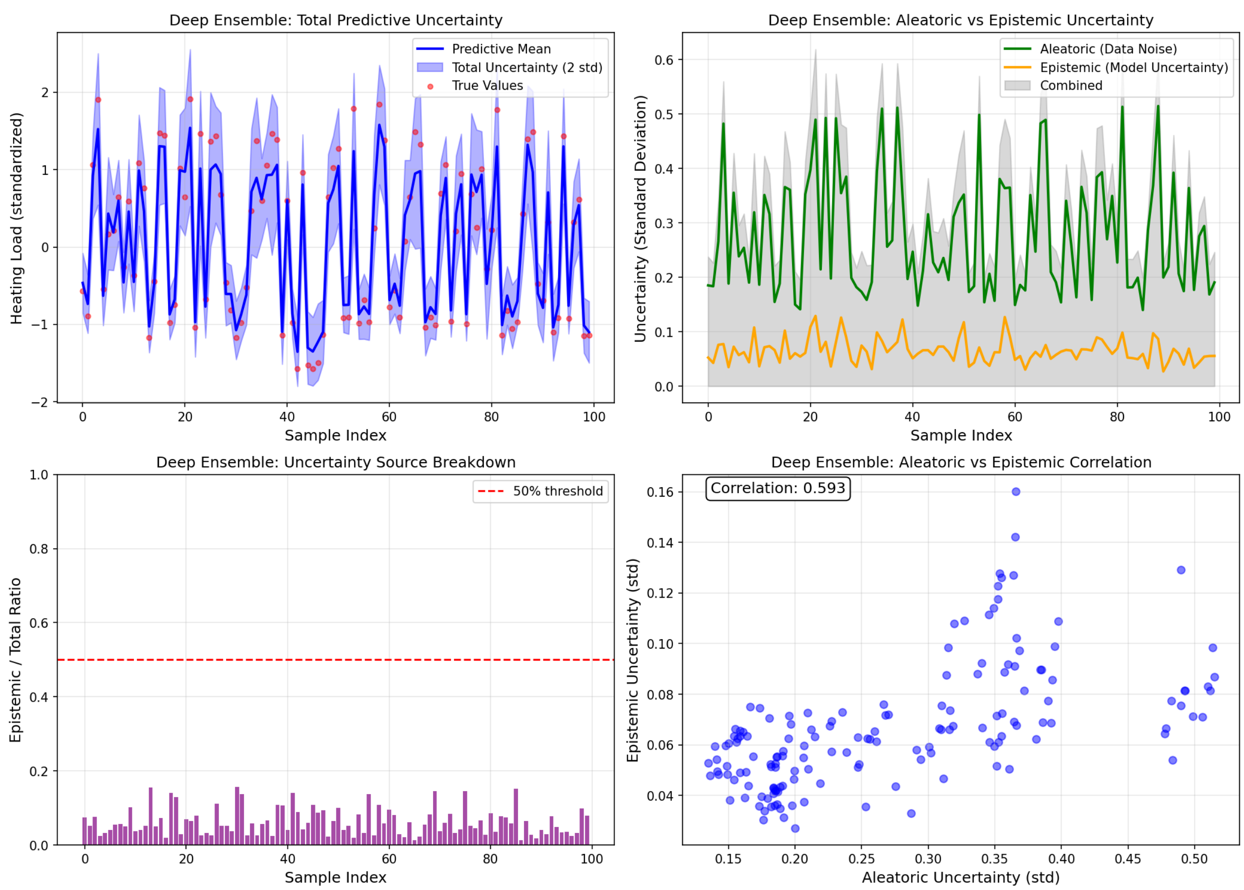
Epistemic vs. Aleatoric Uncertainty

Metric	**MC-Dropout**	**Deep Ensemble**
Uncertainty Components		
Mean Aleatoric uncertainty (std)	0.277693	0.272662
Mean Epistemic uncertainty (std)	0.160979	0.066117
Mean Total uncertainty (std)	0.323618	0.281319
Proportions		
Epistemic proportion	25.22%	6.66%
Aleatoric proportion	74.78%	93.34%

MC-Dropout: total predictive uncertainty, aleatoric vs. epistemic breakdown, epistemic proportion by sample, and aleatoric-epistemic correlation

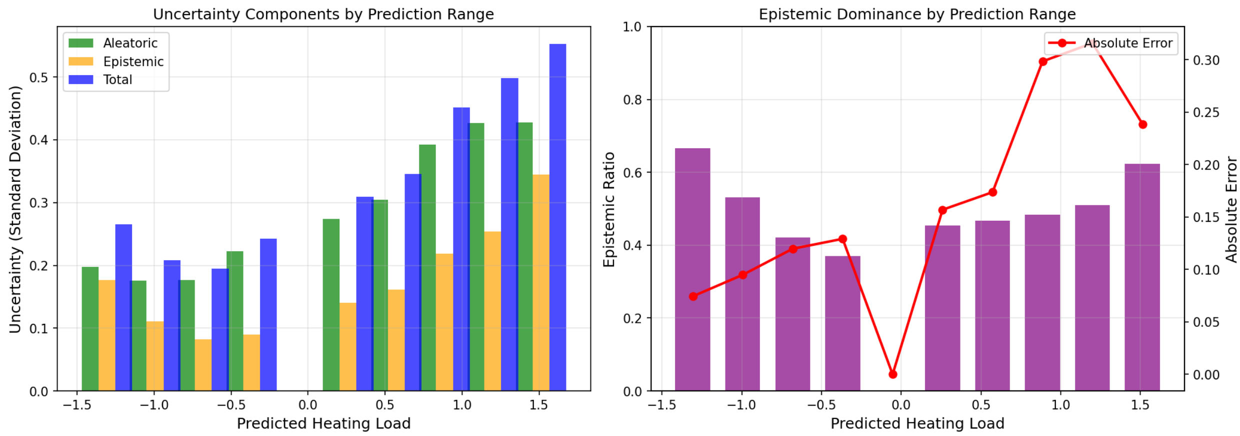


Deep Ensemble: total predictive uncertainty, aleatoric vs. epistemic breakdown, epistemic proportion by sample, and aleatoric-epistemic correlation



Aleatoric uncertainty dominates the total uncertainty for both models. MC-Dropout attributes 74.78% of its total uncertainty to aleatoric and 25.22% to epistemic, while Deep Ensemble attributes 93.34% to aleatoric and only 6.66% to epistemic. MC-Dropout attributes substantially more of its total uncertainty to the epistemic component than Deep Ensemble does. This is a real, interesting difference between the two methods rather than a bug. MC-Dropout's 50 stochastic passes appear to explore a wider range of plausible models than the disagreement between just 5 independently trained ensemble members.

Epistemic Share by Prediction heating-load range



The epistemic share varies with prediction range. At the lowest prediction range, epistemic ratio is 66.62%, while at the highest range it is 62.27%. In the middle range, epistemic ratio drops to as low as 36.96%. The epistemic share is highest at the extremes of the predicted heating-load range and lower in the middle of the range. This is consistent with the standard interpretation that the model is more uncertain about its own knowledge when extrapolating toward the edges of the training data distribution, and relies more on a stable, learned aleatoric noise estimate in the well-sampled middle region.

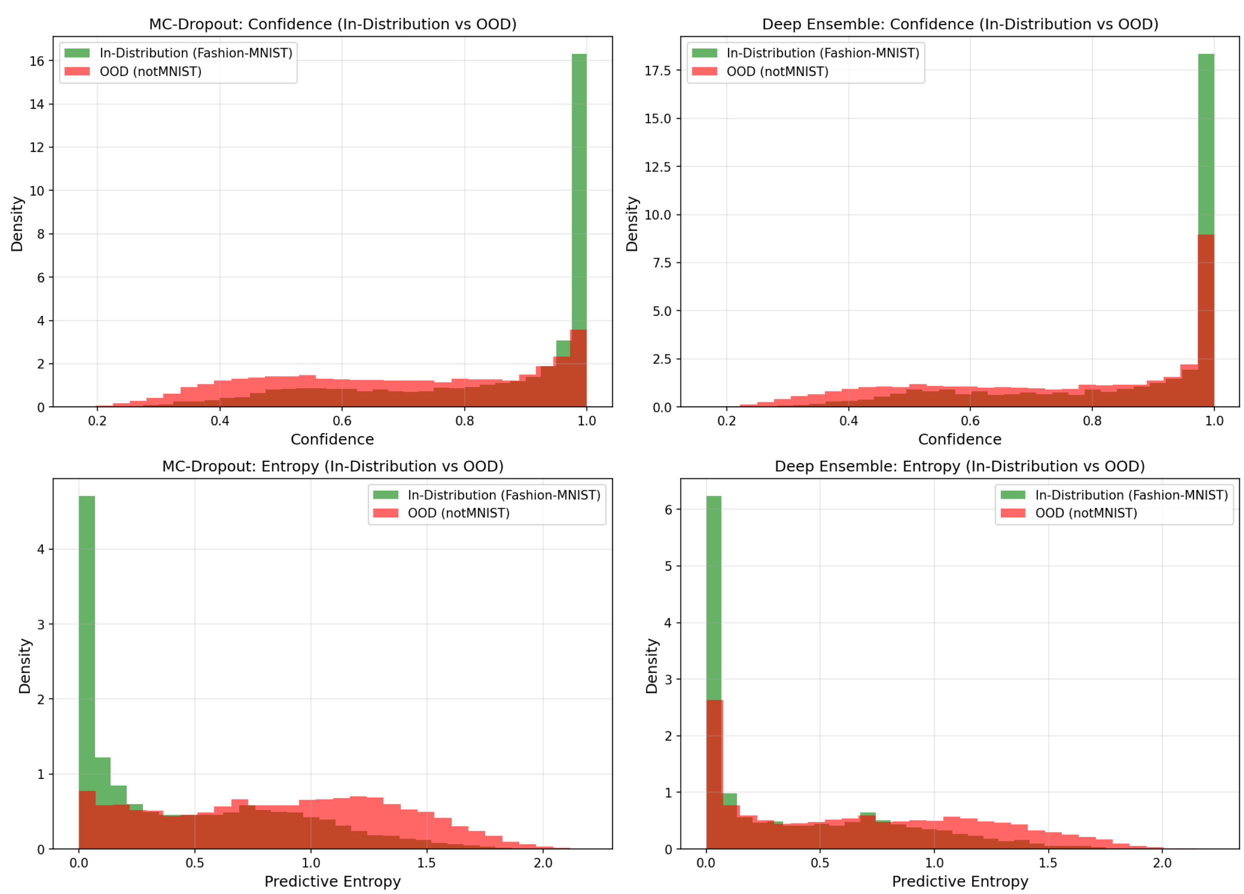
Error magnitude analysis:

Metric	High Error Samples (top 20%, n=31)	Low Error Samples (bottom 20%, n=31)
Mean Aleatoric std	0.422578	0.242076
Mean Epistemic std	0.206473	0.167368
Mean Total std	0.471464	0.295797
Epistemic ratio	43.55%	55.63%

Somewhat counterintuitively, the lowest-error samples had a higher epistemic ratio (55.6%) than the highest-error samples (43.6%). The explanation is visible in the prediction-range breakdown: the highest-error samples in this dataset happen to fall in the middle-to-upper prediction range, where epistemic ratio is only moderate (45–51%), not at the extremes where epistemic ratio peaks. In other words, epistemic uncertainty tracks position in the input space relative to training-data density, not error magnitude directly. The two are correlated but not identical, and aleatoric (irreducible) noise is the larger contributor specifically on the genuinely hardest-to-predict samples in this dataset.

Out-of-Distribution (OOD) Behaviour

Metric	MC-Dropout	Deep Ensemble
Mean confidence	0.6880	0.7458
Mean Entropy	0.8659	0.6792
Confidence std	0.2159	0.2274
Entropy std	0.5056	0.5385



How does uncertainty reflect OOD detection?

Confidence:

On Fashion-MNIST, both models concentrate confidence sharply near 1.0 (green peak). On notMNIST, a completely different domain (letters, not clothing), the confidence distribution (red) flattens out across the full range from ~0.2 to 1.0, with much less mass at the very top. This is the correct qualitative signature of OOD-sensitive uncertainty: the model is markedly less sure of itself on data it was never trained to classify

Entropy:

The same pattern appears in reverse for entropy, sharply peaked near zero for in-distribution data, spread out toward higher entropy values (roughly 0.5–2.0) for OOD data.

Quantitative shifts:

Quantitatively, mean confidence dropped from ~0.84 (in-distribution) to roughly 0.69–0.75 (OOD) for both models, and mean entropy roughly doubled, both directions consistent with the models correctly recognizing notMNIST as unfamiliar, even though they were never explicitly trained to detect OOD inputs.

MC-Dropout showed a somewhat larger relative shift (bigger confidence drop, bigger entropy increase) than Deep Ensemble in my runs, suggesting its dropout-sampling-based uncertainty is slightly more sensitive to distribution shift in this setting than the ensemble-disagreement-based uncertainty.

Important limitation:

Neither model's confidence drops anywhere near 0 on OOD data, a meaningful fraction of notMNIST images still receive confidence above 0.8. This means predictive entropy / confidence alone, without any OOD-specific training, would not be reliable enough on their own to act as a hard OOD detector or rejection rule in this setup; they are a useful soft signal, not a guarantee.

3. Comparative Discussion

When is Uncertainty Reliable?

Uncertainty is most reliable in regions well covered by training data and for genuinely ambiguous/hard inputs (high aleatoric uncertainty correctly reported as high). It is least reliable, in the sense of being only partially informative, at the extremes of the input distribution and on truly out-of-distribution inputs, where both models still retain some residual overconfidence.

Which Model Works Better, and Why?

Criterion	Better model	Basis
Regression accuracy (RMSE)	Deep Ensemble	2.2127 vs 2.3853
Regression likelihood (NLL)	Deep Ensemble	-1.0933 vs -1.0196
Regression calibration (ECE)	Deep Ensemble	0.0419 vs 0.0631
Classification accuracy	Deep Ensemble	88.12% vs 87.55%
Classification likelihood (NLL)	Deep Ensemble	0.3366 vs 0.3520
Classification calibration (ECE)	Deep Ensemble	0.0277 vs 0.0371
Training compute	MC-Dropout	trains 1 model vs. 5
Inference compute	Deep Ensemble	5 forward passes vs. 50

Deep Ensemble outperforms MC-Dropout on every accuracy and calibration metric we measured, on both datasets. This matches the pattern reported in the literature (Lakshminarayanan et al., 2017): training several independently-initialized networks and averaging tends to produce both better point predictions (an ensembling effect familiar from classical ML) and better-calibrated uncertainty than a single dropout-regularized network sampled stochastically at test time.

Computational Trade-offs

Aspect	MC-Dropout	Deep Ensemble
Training time	1× (one model)	~5× (five independent models)
Stored parameter sets	1	5
Inference time per prediction	High (50 forward passes)	Moderate (5 forward passes)
Parallelizability of inference	Low (sequential passes through the same model, though batchable)	High (5 independent models, trivially parallel)
Memory footprint	Low (one model)	High (5 models)

MC-Dropout shifts cost to inference time, Deep Ensembles shift cost to training time and storage. For a deployment scenario with many inference requests and infrequent retraining, Deep Ensembles' cheaper, parallelizable inference is usually preferable despite the higher training cost. For rapid prototyping or memory-constrained deployment, MC-Dropout is more practical.

Out-of-Distribution Behaviour

We evaluated both Fashion-MNIST-trained classifiers on notMNIST (images of letters, a completely different domain) as an OOD test. The full plots and discussion of how uncertainty reflects OOD detection are presented above, below is the comparative angle relevant to "which model works better."

- Both models correctly show lower confidence and higher entropy on notMNIST than on Fashion-MNIST.
- MC-Dropout showed the larger relative drop in confidence and rise in entropy on OOD data in our runs, suggesting its stochastic-mask-based uncertainty is somewhat more sensitive to distribution shift than the ensemble-disagreement-based uncertainty in this setup.
- Neither model's OOD confidence drops low enough to act as a reliable standalone OOD detector, this is a shared limitation of both methods rather than an advantage of either.

4. Reflections

What I Learned

1. Standard neural networks give no signal about their own uncertainty, they can be just as "confident" on nonsense inputs as on inputs they were trained for. Both Bayesian-DL approximations we implemented largely fix this, to differing degrees.
2. Total predictive uncertainty is not a single quantity , separating it into aleatoric and epistemic tells us what kind of action would actually help, which a single combined uncertainty number cannot.
3. Calibration is a separate axis from raw accuracy, a model can be accurate but overconfident, or less accurate but well-calibrated. ECE is a useful tool for measuring this directly rather than relying on accuracy alone.
4. Deep Ensembles' advantage in my experiments came at a clear, predictable 5× training cost, so the better-performing method is also the more expensive one to train.

Limitations

- No systematic hyperparameter search was performed; results may not represent each method's best achievable performance.
- As k =5 larger ensemble would likely give a smoother, more reliable epistemic uncertainty estimate, at proportionally higher training cost.

- There was relatively small UCI dataset (~770 samples total) and this limits how confidently the regression findings generalize to other tabular problems.
- Neither model was trained with any mechanism (e.g. outlier exposure) specifically intended to suppress confidence on OOD inputs, which likely explains the residual OOD overconfidence observed.

Future Ideas

- We can combine the two approaches, an ensemble of MC-Dropout models, to capture both sources of stochasticity simultaneously.
- Apply post-hoc calibration (e.g. temperature scaling) on top of either method to further reduce ECE without retraining.
- Replace the Fashion-MNIST MLP with a small CNN to test whether the uncertainty-quality findings hold with a stronger base architecture.
- Using epistemic uncertainty estimates to drive an active learning loop, preferentially labeling/training on samples the model is most epistemically uncertain about..